PROJECT TITLE:

AI-DRIVEN FOOD SURPLUS DONATION SYSTEM FOR ERADICATING HUNGER

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1. Abstract

Food waste is a critical global challenge with social, economic, and environmental ramifications. According to the Food and Agriculture Organization (FAO) of the United Nations, the world consumes around 1.3 billion tonnes of food annually— but nearly one-third goes to waste or is lost in production processes. This disparity means that millions go hungry each day, predominantly in developing nations where malnutrition is also prevalent among those who do have access to food supplies.

To bridge this gap between abundance and need, our aim is to implement an **AI-driven food surplus donation system** that connects businesses with surplus food to local charities and food banks. This platform will facilitate food redistribution, reduce waste, and maximize the impact on hunger alleviation, ultimately contributing to a more sustainable and equitable food system.

2.Introduction

Efficient waste management strategies are one of the key elements for Sustainable Development. Food waste has been identified as a priority waste stream due to its negative social, economic and environmental impacts. International commitments to reduce waste were included in the 2030 Agenda for Sustainable Development with the most favorable option being **prevention.** However, considering that there will always be a considerable amount of food waste that is unavoidable, effective management strategies are also needed to ensure that surplus food acceptable for human consumption is actually used to feed people.

One effective approach is food donation. Although the concept of donation traces back to the thirteenth century, its modern implementation has proved fruitful. It not only reduces food wastage but also has the potential to positively influence food security by improving access to food, as encouraged by Food and Agriculture Organization (FAO) of the United Nations.

With this in mind, our AI**-driven food surplus donation system aims** to act as a bridge between those seeking to dispose of surplus food and those in need of food.

3.Problem Statement

A significant amount of food is wasted annually while millions of people remain undernourished. Restaurants, markets, bakeries, and households regularly dispose of edible food due to overproduction or approaching expiry dates. Existing donation systems are fragmented, lack automation, and often fail to ensure food reaches the right recipients in time. The lack of real-time data, predictive analytics, and optimization in food donation logistics has created a gap that this system aims to fill.

4.Objectives

- **Develop an AI-based system** that matches donors with surplus food to those in need(recipients)

- **Design a web interface** that is responsive to the two roles : donors and recipients (those in need)

- **Reduce food waste** by improving the speed and efficiency of redistribution.

5.Methodology

Our system follows a data-driven approach to match food surplus donations to suitable recipients using machine learning. The methodology is broken down into the following phases:

5.1 Data Collection and Preprocessing

- **Donor Data**: Collected from an .ods file, including fields like food_type, expiry_date, surplus_quantity, city, and geographical coordinates (lat, lng) mapped to the cities. Only non-expired donations were retained.

- **Recipient Data**: Simulated records include recipients' food preferences (multi-selection), city, and required quantity.

- **Normalization**:
  All food_type values were lowercased and stripped of whitespace for consistency.
  Cities were title-cased and mapped to coordinates for distance calculations.

5.2 Feature Engineering

From each **donor-recipients** pair,we generated key features to quantify compatibility :

Feature	Description
Food match	**Binary : 1** if donor's food type in recipient's food preference list
Distance	Geographical distance (in km) between donor and recipient using the Haversine method (via geopy).
Exact quantity match	**Binary : 1** if donor's surplus meets or exceeds recipient's need

Quantity Ratio Continuous : Ratio of surplus to need capped at 1.0

CONCLUSION : These features were saved along with a binary `label`: **1** if the donation is a viable match (food match + nearby + available surplus), else **0**.

5.3 Label Generation

Each donor was paired with all recipients to create a labeled dataset of possible matches.The following logic was used to assign labels to our data :

- **food_match**: Set to **True** if the donor's food type is present in the recipient's food preferences (which is a list, as recipients can request multiple food types).
- **distance**: Calculated using the Haversine formula (**geopy** library) based on latitude and longitude of both donor and recipient.
- **surplus_quantity**: Donor's available quantity of food.

Match Criteria

A match (label = 1) was assigned if all of the following conditions were met:

- food_match == True
- distance ≤ 50 km
- surplus_quantity > 0

Otherwise, the label was set to 0 (non-match).

The result was a comprehensive labeled dataset representing **positive and negative match instances**, saved to **labeled_matches_finale2.csv**

Food Type Encoding and Categorization

To support robust matching and future scalability:

- Donor food types were encoded using **LabelEncoder,** converting each food type into a numeric category. This ensures efficient storage and compatibility with ML models, especially for clustering or collaborative filtering later.
- Recipient food types remained as lists of strings, allowing multi-selection and filtering logic during the matching process.
- A master list of all unique food types (from both donors and recipients) was compiled and saved as **all_food_types_list_finalee2.pkl** to maintain encoding consistency across models and future data inputs.

This structure supports:

- **Flexible matching**: A single donor food type can be matched to any number of recipients requesting it.

- **Extensibility**: New food types can be added without breaking the system, using the saved list for retraining.

This structured labeling ensures the model learns to distinguish real, viable donation opportunities from impractical ones. It sets the foundation for high-quality recommendations that prioritize recipient need, donor capability, and logistical feasibility.

5.4 Model Training

- Random Forest Classifier was used for our tabular data to handle non-linear decision boundaries.

- Input features: **food_match, exact_quantity_match, quantity_ratio**, and **distance**.

- Target: **label** (1 = good match, 0 = not a match).

- Dataset Split: 80% training, 20% testing.

- Our trained model was trained and saved as **match_model_finalee2.pkl**.

5.5 Model Prediction

To predict whether a new donor-recipient pair is a good match: ● A

feature vector is constructed using the same logic as training.

- The model outputs 1 (match) or 0 (no match).

- This prediction can trigger real-time notifications to both parties.

Sampled Input Example :

```
{
'food_match': 1,
'exact_quantity_match': 0,
d 'quantity_ratio': 0.5,
'distance': 12.5
}
```

5.6. Model Persistence

Models and encoders were persisted using **pickle** and **joblib** for reuse in the system's API or real-time engine.

Saved files:

- **match_model_finalee2.pkl** – trained RandomForest model

- **food_encoder_donor_finalee2.pkl** – label encoder for food types

- **all_cities_finalee2.pkl, all_cities_tocoords_finalee2.pkl** – for city-coordinate mapping

- **all_food_types_list_finalee2.pkl** – complete list of food types for consistency

This methodology ensures:

- **Scalability:** New recipients/donors can be matched in real-time using the saved model.

- **Fairness**: Claims are match-based rather than manual or biased.
- **Flexibility**: Supports multi-food preferences, location-based filtering, and future upgrades to collaborative filtering models.

6.System Architecture

Frontend : React + Vite
Backend : Django + Django Rest Framework
Database : PostgreSQL
Websocket Communication : Django channels + redis
AI/ML Algorithms : Greedy Matching and Collaborative filtering
Key Components :

User Roles : Donor,Recipient,Admin

Donation Flow :

- Donor adds donation
- AI matches donation to recipients
- Recipient chooses to either claim donation or not
  - However if recipient claims then both roles notified of the match

AI Matching Logic :

- Greedy Algorithm : Matches based on need,availability and proximity

7.Features

- **Donation Submission** : Donor submits available food donation with its expiration date,quantity alongside details(optional) of pickup date and location
- **Matching engine** : Smart suggestions based on need,availability and location
- **Real time Notifications** : Both donor and recipient notified of the matches
- **Claim-based system** : Gives all eligible recipients a fair chance to access and claim donations based on when they respond—removing bias and manual selection.

8.Technology Used

Technology	Purpose
React + Vite	Frontend framework for building UI
Django and Django Rest Framework	Backend web framework for handling APIs and logic
PostgreSQL	Relational Database for persistent data storage
Tailwind CSS Radix Sweet Alert2	Styling

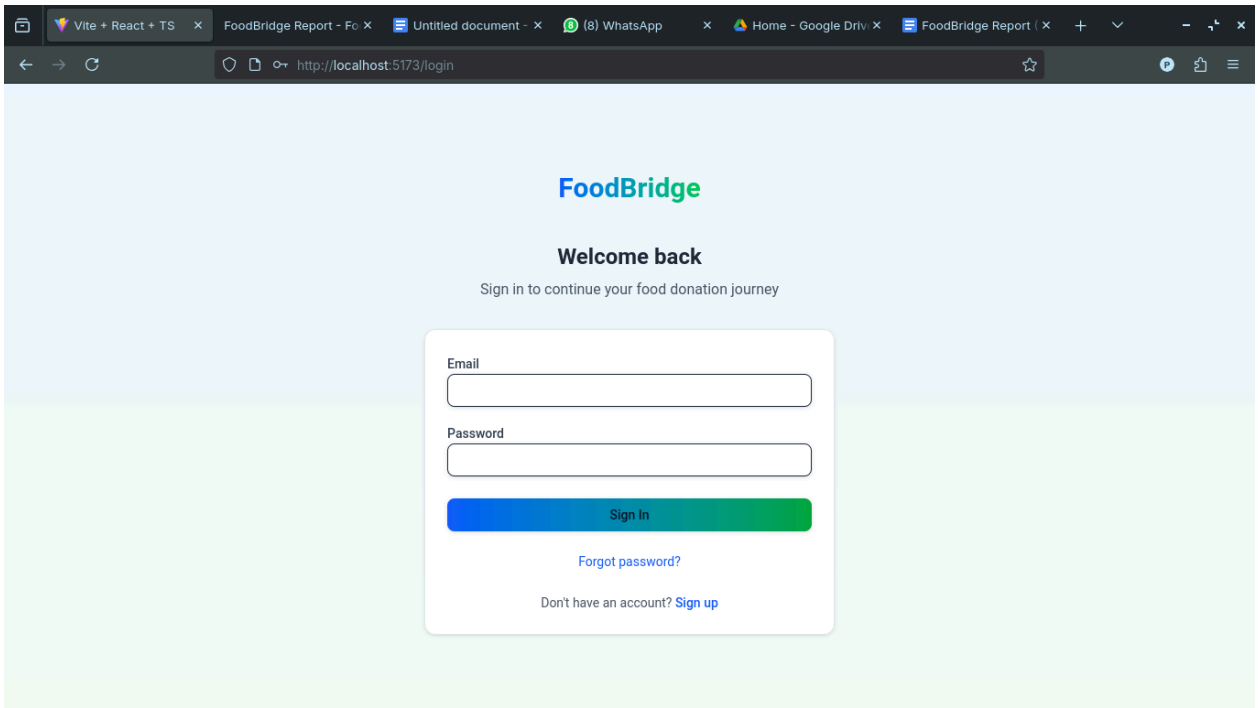
CSS	
Django channels	Real time websocket notifications

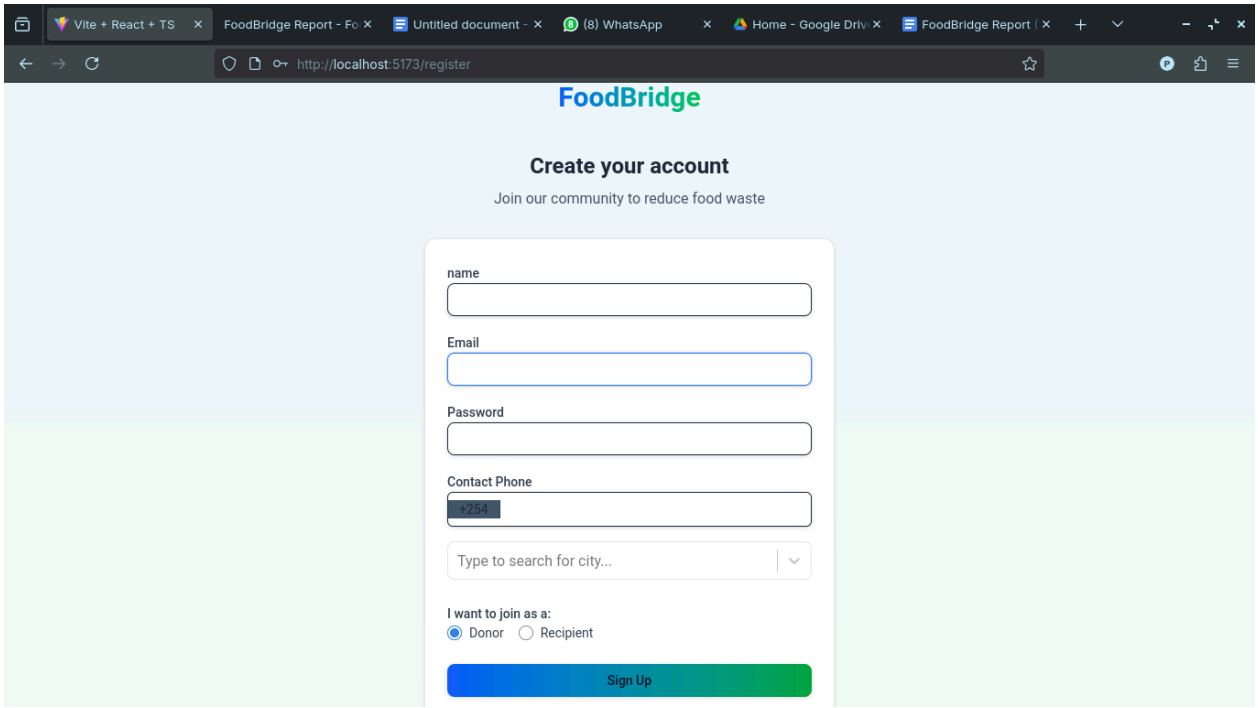
Machine Learning (Python) Matching Algorithm -> Donor-Recipient Matching -> Greedy Algorithm

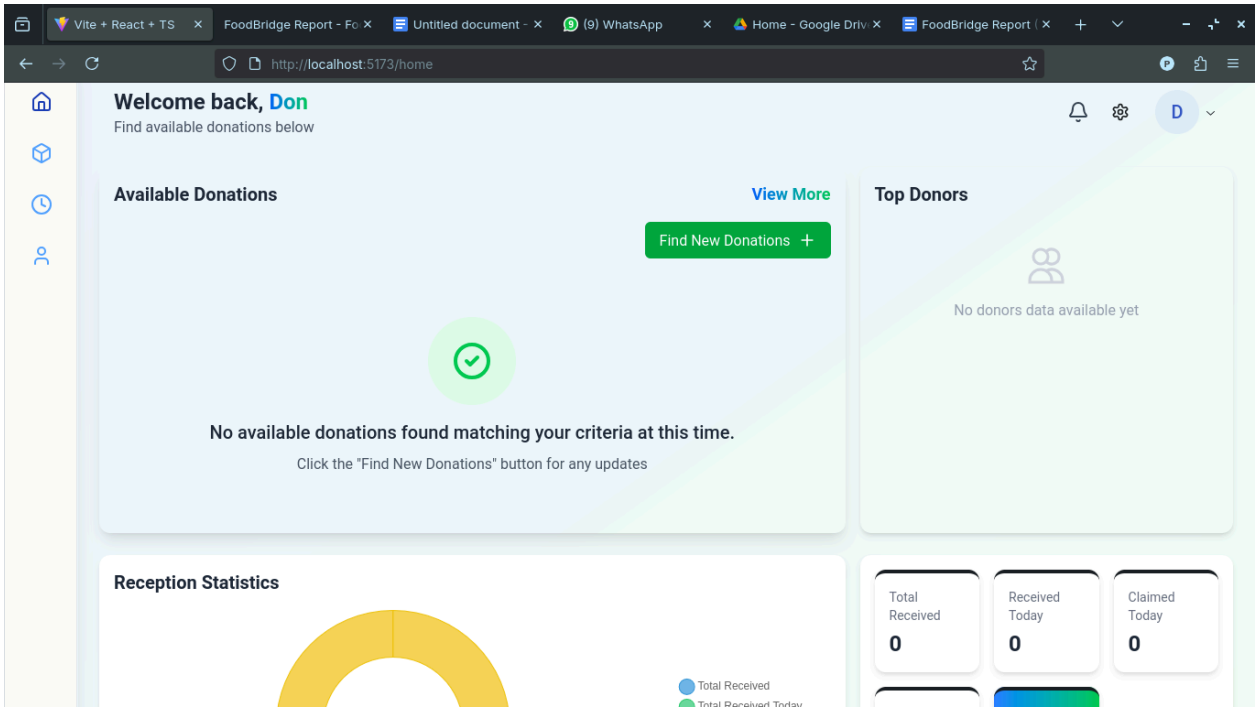
Redis Message broker for real time notifications

9.Screenshots

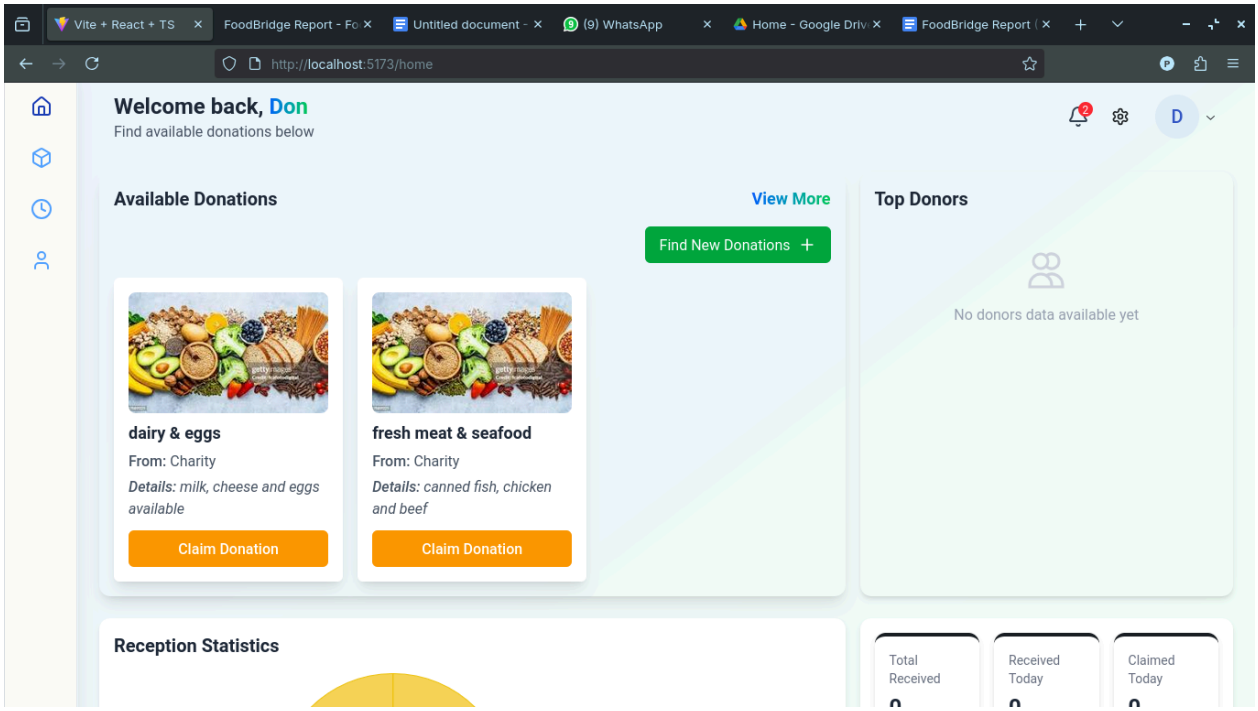
- ## Authentication page



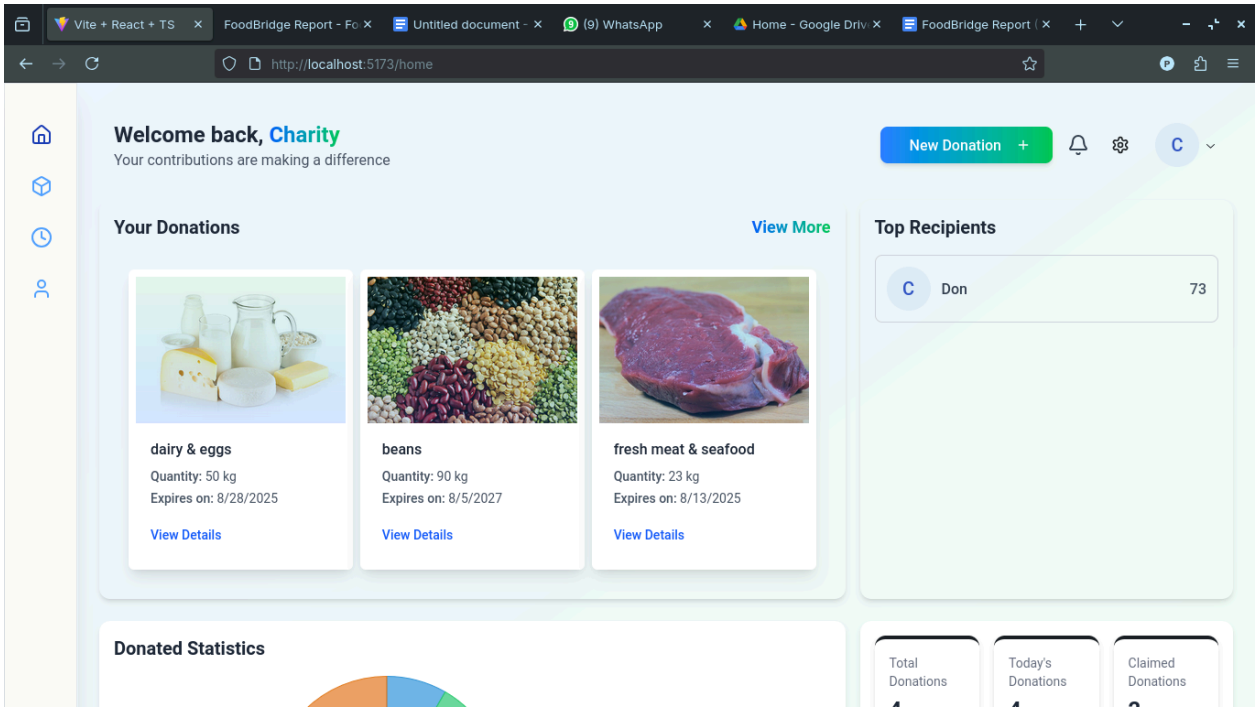

Vite + React + TS
FoodBridge Report - Fo
Untitled document -
(8) WhatsApp
Home - Google Driv
FoodBridge Report (
http://localhost:5173/register
FoodBridge
Create your account
Join our community to reduce food waste
name
Email
Password
Contact Phone
+254
Type to search for city...
I want to join as a:
Donor
Recipient
Sign Up



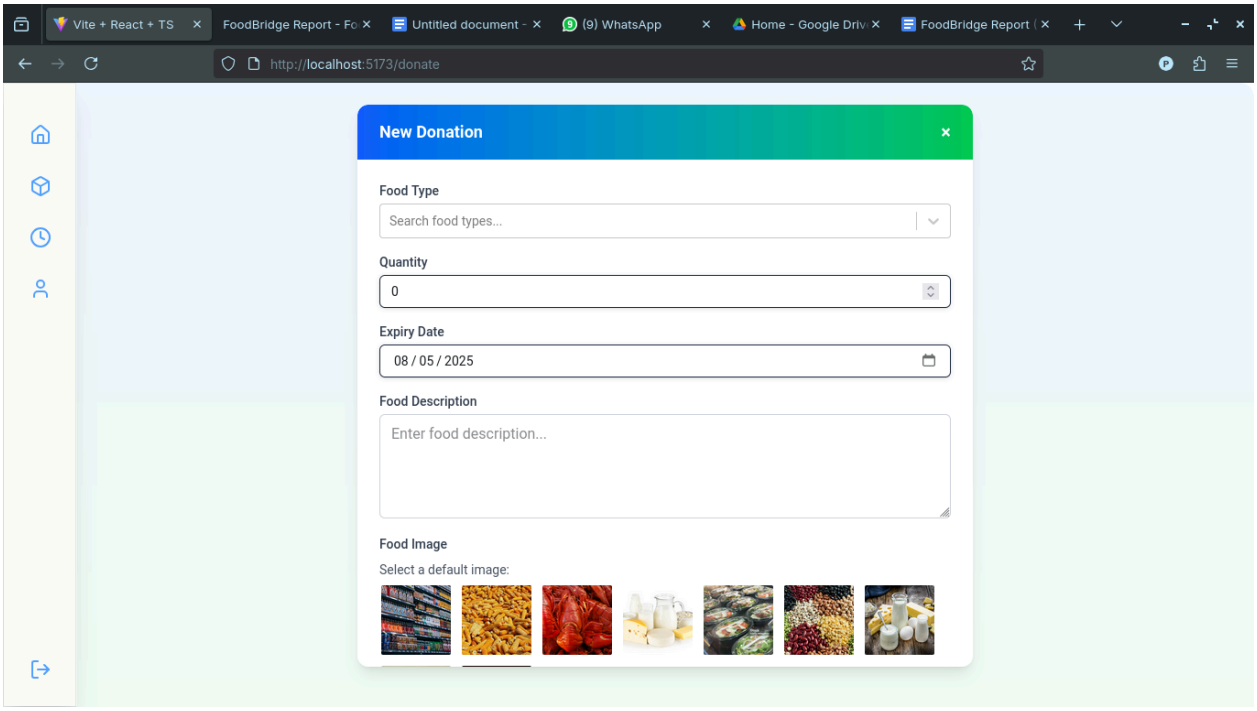
- **Recipient's dashboard**



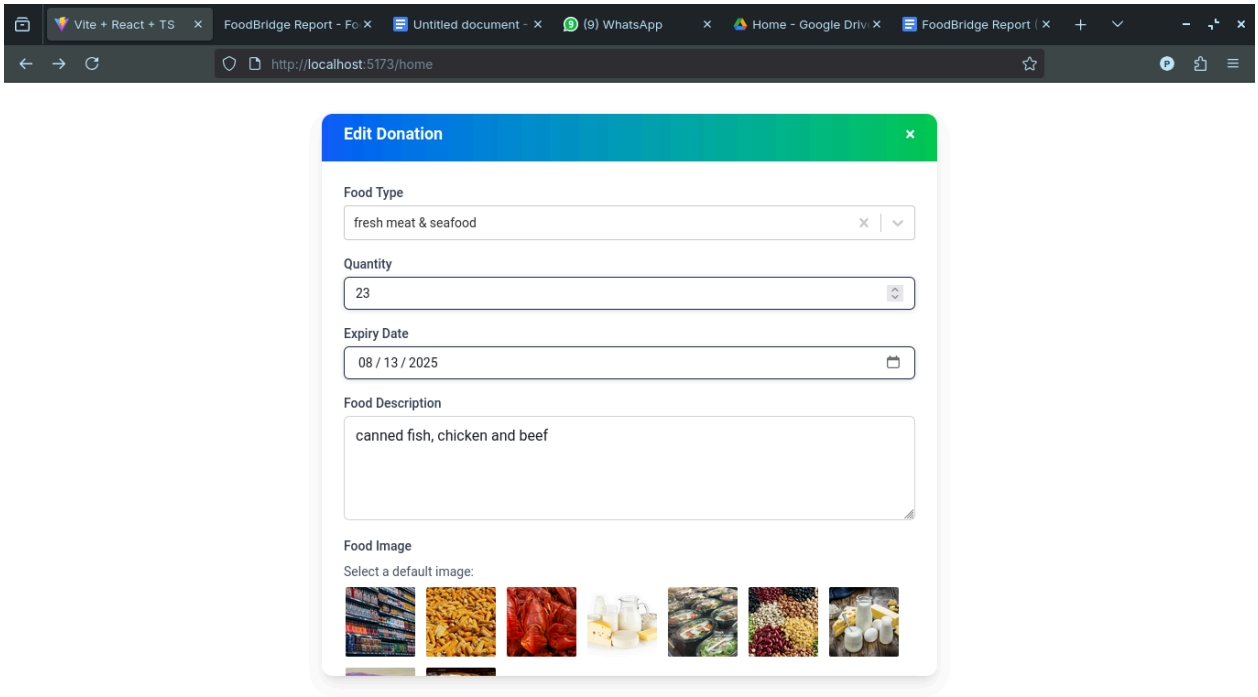
- **Recipient's page with donation match**

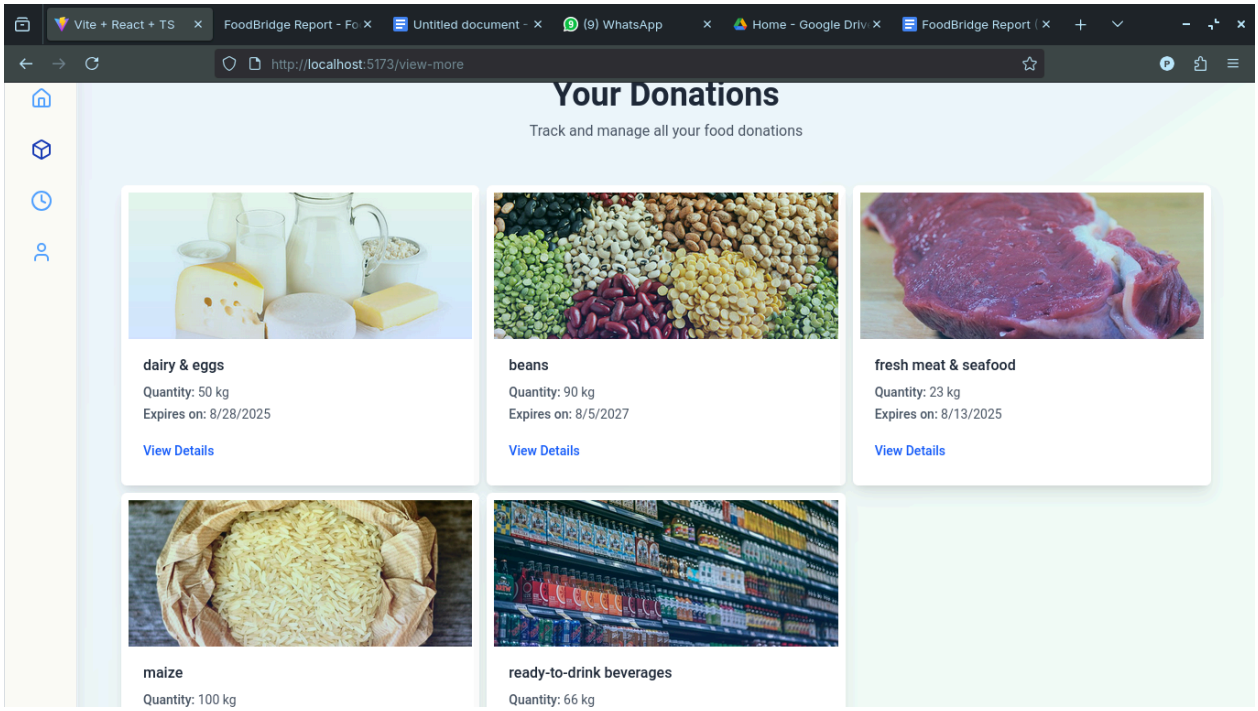


- **Donor's dashboard**

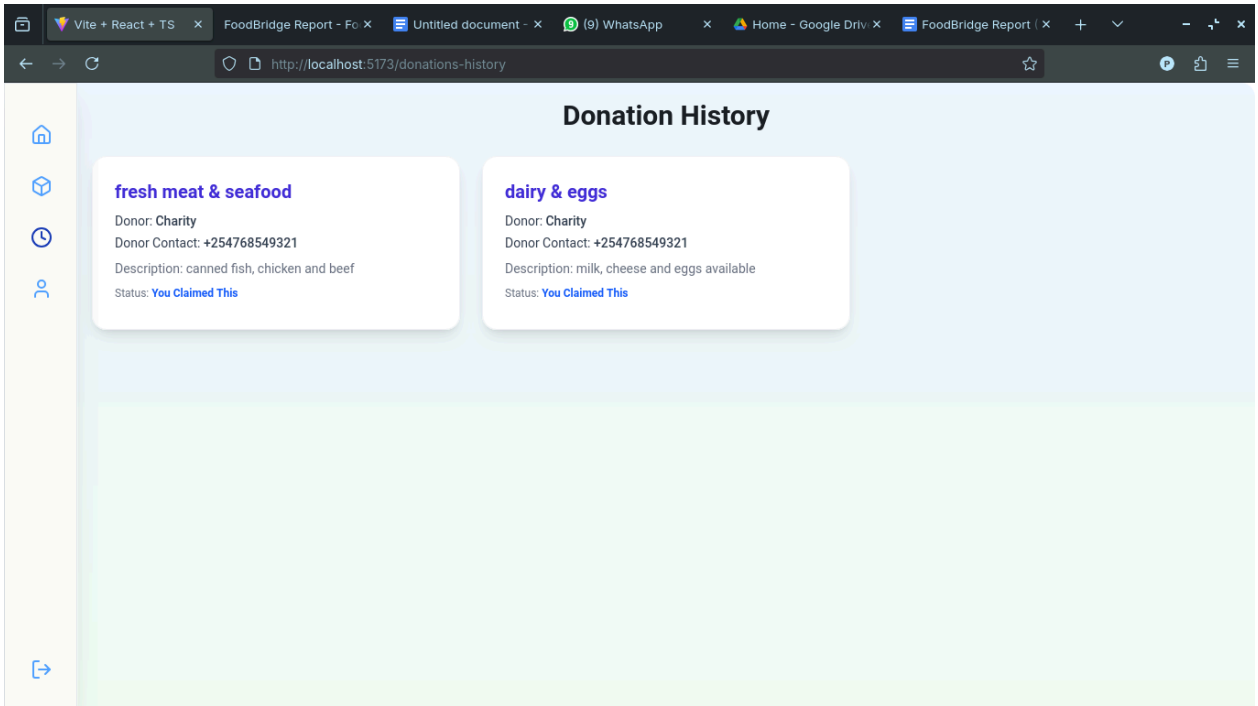
- 

- **New donation modal**

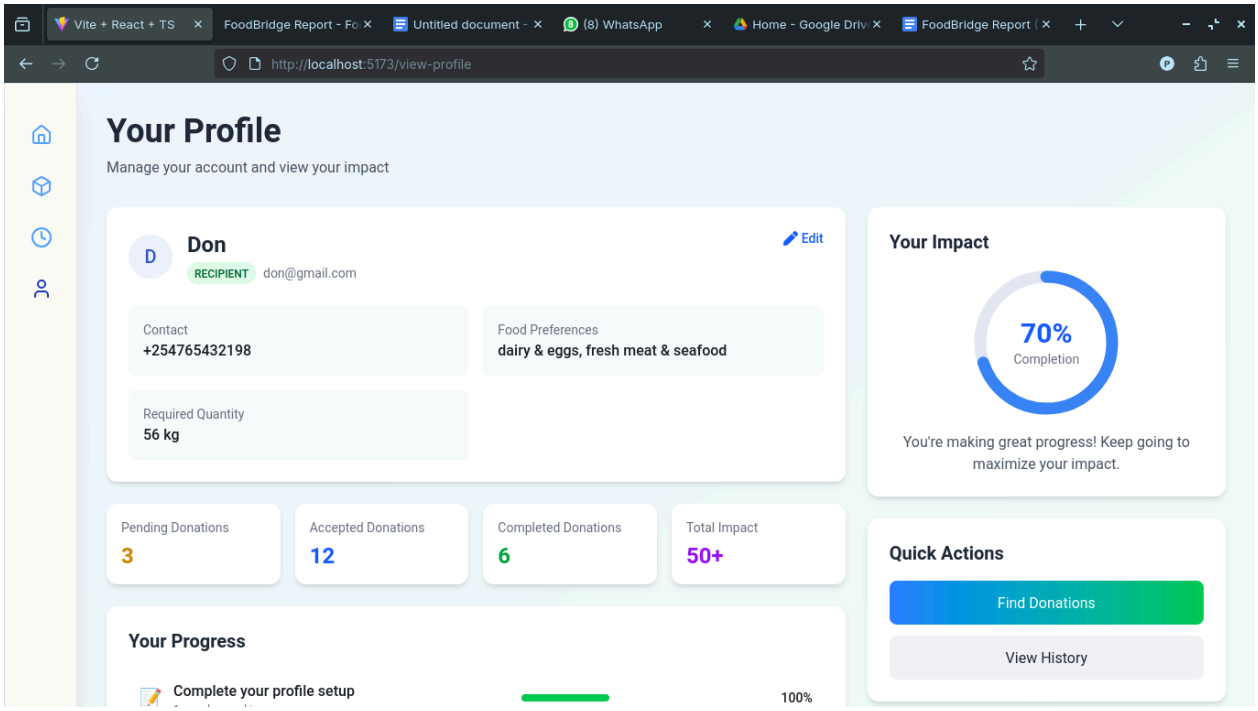




- **All Donations page**



- **Donation history page with donation status**



- **User profile page**

10.Challenges faced

- Balancing UX for both donor and recipient flows
- *Real-time performance:* Ensuring low latency between donation and match notification using Django Channels and Redis required tuning server configs and message queue settings.
- Handling edge cases in AI recommendations like fine tuning feature weights (need,availability and location) was needed
- Deployment

11.Conclusion

This system has explored the design and development of an AI-driven food surplus donation system aimed at addressing the persistent challenges of food waste and food insecurity. By applying artificial intelligence and intelligent matching algorithms, the system facilitates efficient redistribution of surplus food from donors to recipients, ensuring that available resources are utilized optimally and reach those in need in a timely manner.

The platform integrates key technological components — including machine learning-based recommendation systems, real-time data processing, and an intuitive user interface — to create a solution that is both scalable and adaptable to varying operational contexts. The architecture supports automated decision-making, improves coordination between stakeholders, and minimizes logistical inefficiencies. This project reinforces the critical role of AI in solving real-world societal challenges, particularly in promoting responsible consumption and achieving zero hunger.

12.Future Work

- Implement geospatial routing for pickup optimization
- Extend to mobile apps using acceptable technology stacks
- Add user feedback and rating systems
- Improve ML models with more training data
- Partner with logistics companies for delivery automation
- Completion status for tracking donations upon delivery
- Introduce Achievements and Gamification system features

  Reward users (both donors and recipients) with badges or milestones for consistent participation, donation impact, timely pickups, etc., to boost engagement and encourage positive behavior on the platform.

13.References

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